BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Quiz 3 Duration: 20 min
Semester: Summer 25 Full Marks: 15

CSE 470: Software Engineering

Name: **ID:** **Section:**

Set-*

A weather monitoring system collects data from various environmental sensors, including temperature, humidity, and atmospheric pressure. These sensors are placed across different locations, collecting real-time data at varying intervals. The data collected is then displayed on multiple screens, such as a web-based dashboard, a mobile app, and an automated alert system that notifies users when certain thresholds are met (like extreme temperature or humidity levels). The system is designed to update all displays automatically whenever there is a change in the sensor data, ensuring real-time information is visible to users across different platforms. To manage this, the system follows a centralized configuration system, where only one central configuration object handles and manages the frequency of sensor data collection, update intervals, and the communication with the sensor network. This configuration object is globally accessible across the entire system, ensuring consistency in how data is updated, and that no multiple conflicting configurations are created.

1	**CO1**	**a. Which design pattern/patterns are used here ?Why [5]** **b. Write down the code of the scenario based on your previous answer ? [10]**	**15**

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In a smart home system, various devices such as lights, thermostats, security cameras, and door locks are integrated into a single network. Each device has its own unique communication protocol, but the system needs to provide a unified interface for users to interact with them seamlessly. To manage this, the system uses a centralized manager that controls the settings for all devices. This manager is responsible for adjusting the temperature settings for thermostats, security camera preferences, and lock/unlock schedules for the door locks. The configuration manager ensures that there is only one instance responsible for managing all device settings across the entire system. This ensures consistency and prevents any conflicting settings from being applied. Since the devices use different communication protocols—some rely on Zigbee, others on Wi-Fi, and some use Bluetooth—the system uses a mechanism that translates each device's protocol into a common protocol. This ensures that the centralized manager can communicate with all devices in a uniform way, regardless of the protocol each one uses.

1	**CO1**	**c. Which design pattern/patterns are used here ?Why [5]** **d. Write down the code of the scenario based on your previous answer ? [10]**	**15**